AGING AT THE VERY TOP

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Aging at the Very Top

Chief Executive CEO INSIGHTS EVENTS MAGAZINE NEWSLETTERS

American CEOs Appear to be Getting Older

Ross Kelly

Source: chiefexecutive.net

Aging at the Very Top

BUSINESS INSIDER

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America's corporate gerontocracy

Joe Biden stepped aside. A lot of aging CEOs should do the same.

Source: Business Insider

CEOs Are Getting Older & That's Concerning

Source: Cowen Partners Executive Search

What We Do

- How have age structures among CEOs changed?
- How have CEOs' career paths evolved?
- Which market features can explain these trends?

What We Do

- How have age structures among CEOs changed?

  Document development of CEO age across firms and over time

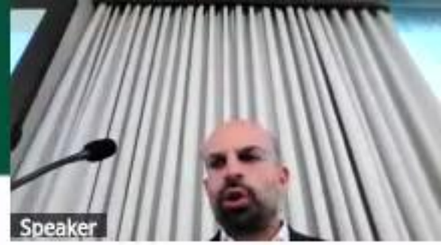
- How have CEOs' career paths evolved?

  Use granular data on CEOs' employment histories

- Which market features can explain these trends?

  Explore factors that have contributed to older CEO appointments since 2000

What We Find



- CEO age at appointment increased substantially since 2000 (7-8 years)
  - $> 3\times$ (college-educated) labor force
- Career paths of prospective CEOs have changed
  - **External experience and labor market mobility** $\uparrow$

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  $\rightarrow$ Interpretation: accumulation of **generalist skills** Alternative Explanations
- Causal analysis and quantitative decomposition
  - Demand-side factors the largest driver of CEO aging trend
  - Substantial contributions of the supply side and its interaction with demand

Main Data Sources

BoardEx

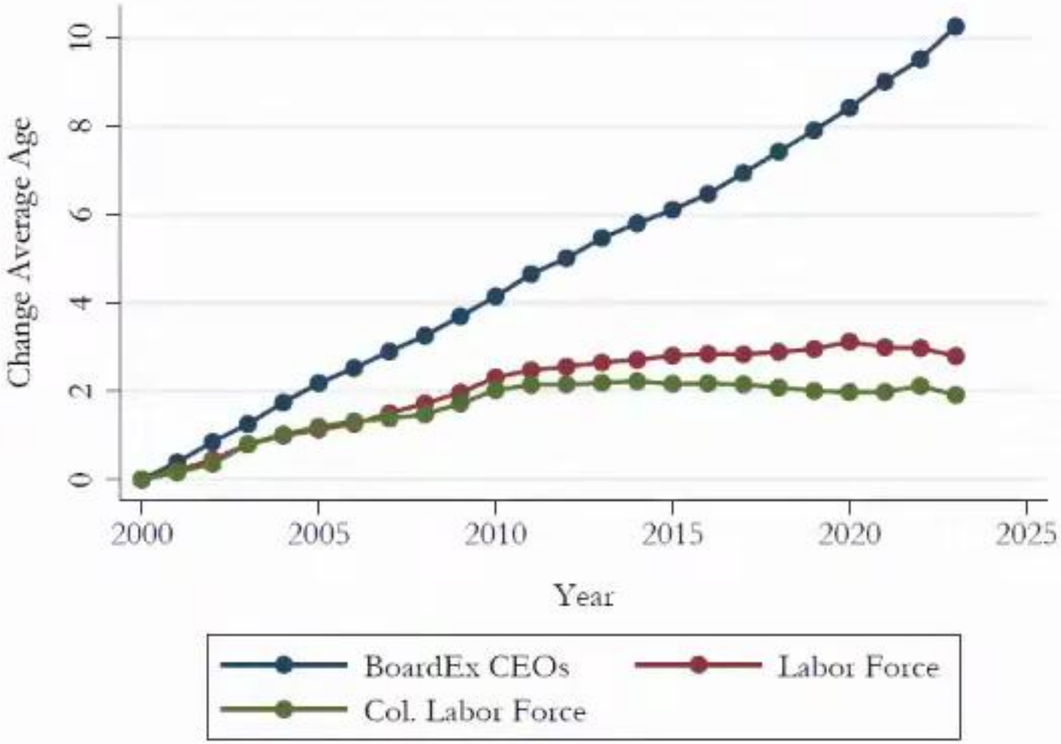
- ≈ 51k CEOs in the US (since 2000)
- Information on executives: age, employment histories, etc.

LinkedIn Data (via Revelio Labs)

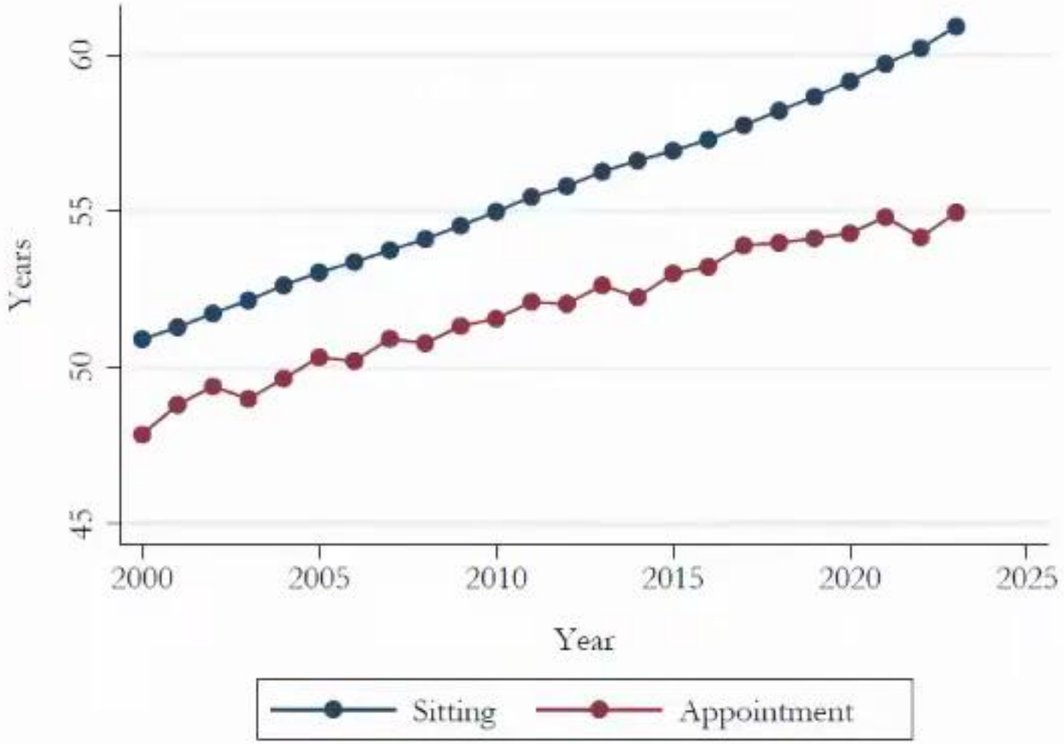
- ≈ 450 MN labor market biographies worldwide
- Identifiers for MSAs, NAICS-4 sectors, and seniority levels
- Combined with BoardEx at firm level using fuzzy merge algorithm

CEO Age Over Time

(a) CEOs vs. Labor Force

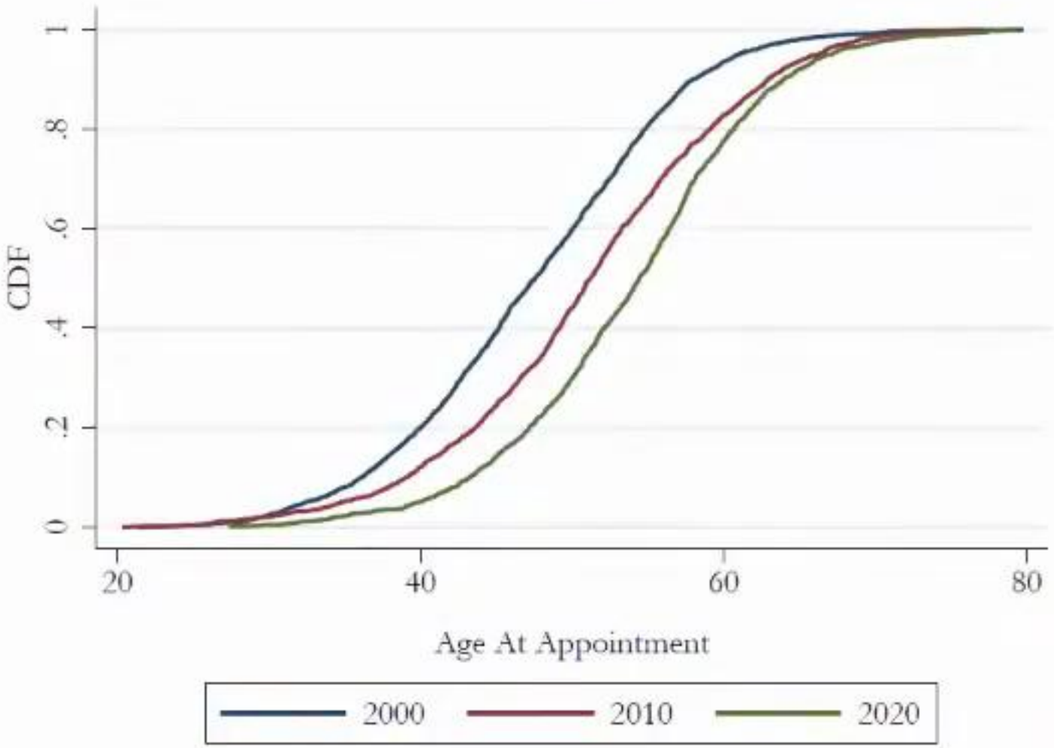


(b) Sitting vs. At Appointment

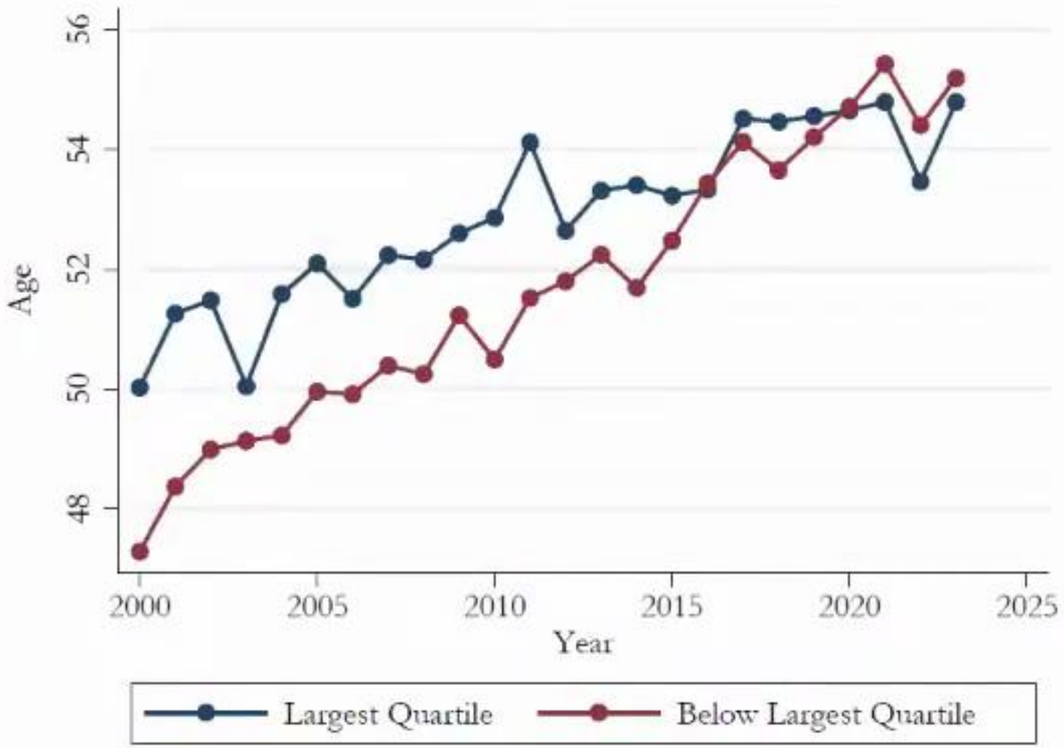


CEO Age At Appointment Over Time

(a) CDF By Year

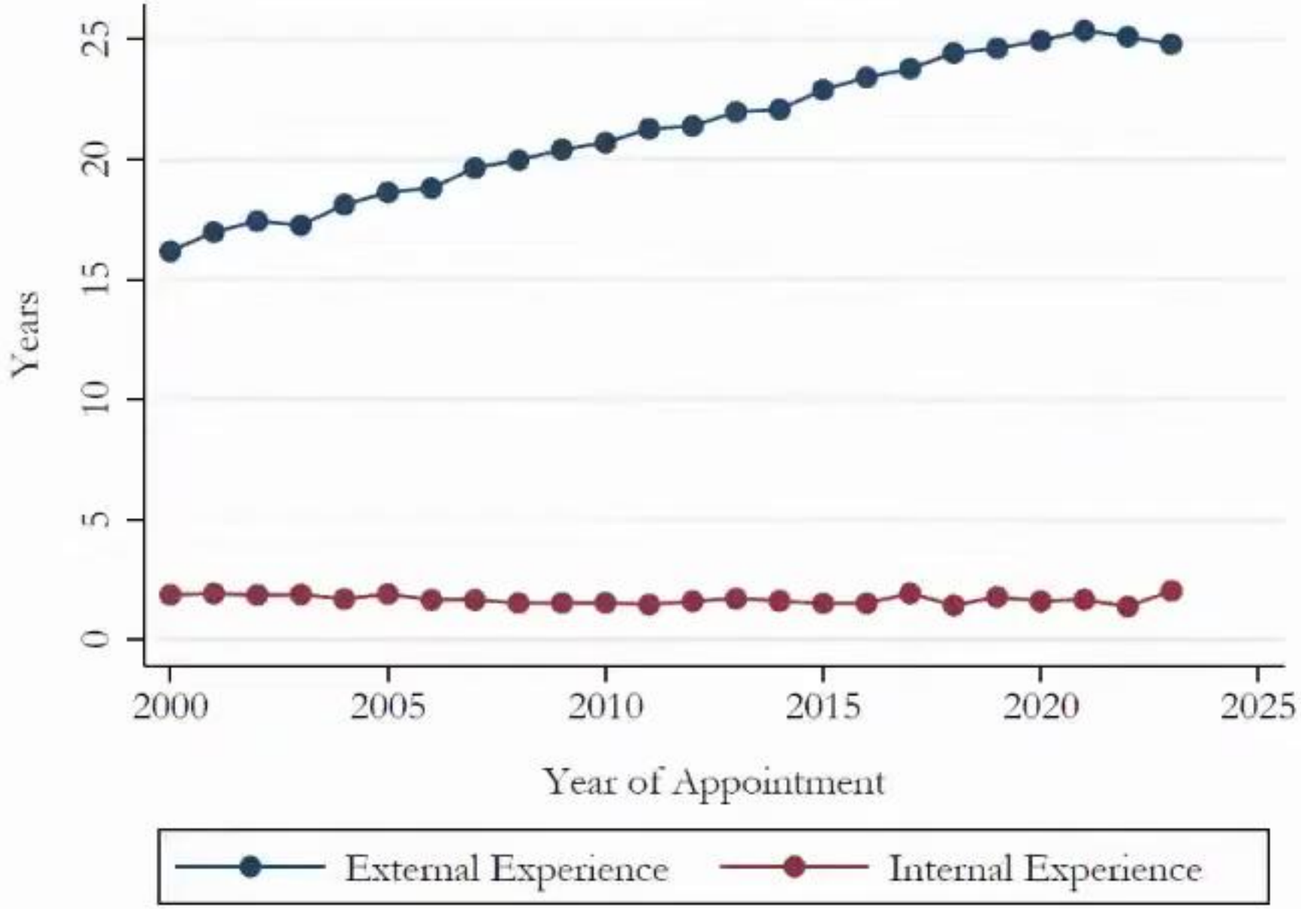


(b) By Firm Size (Number of Employees)

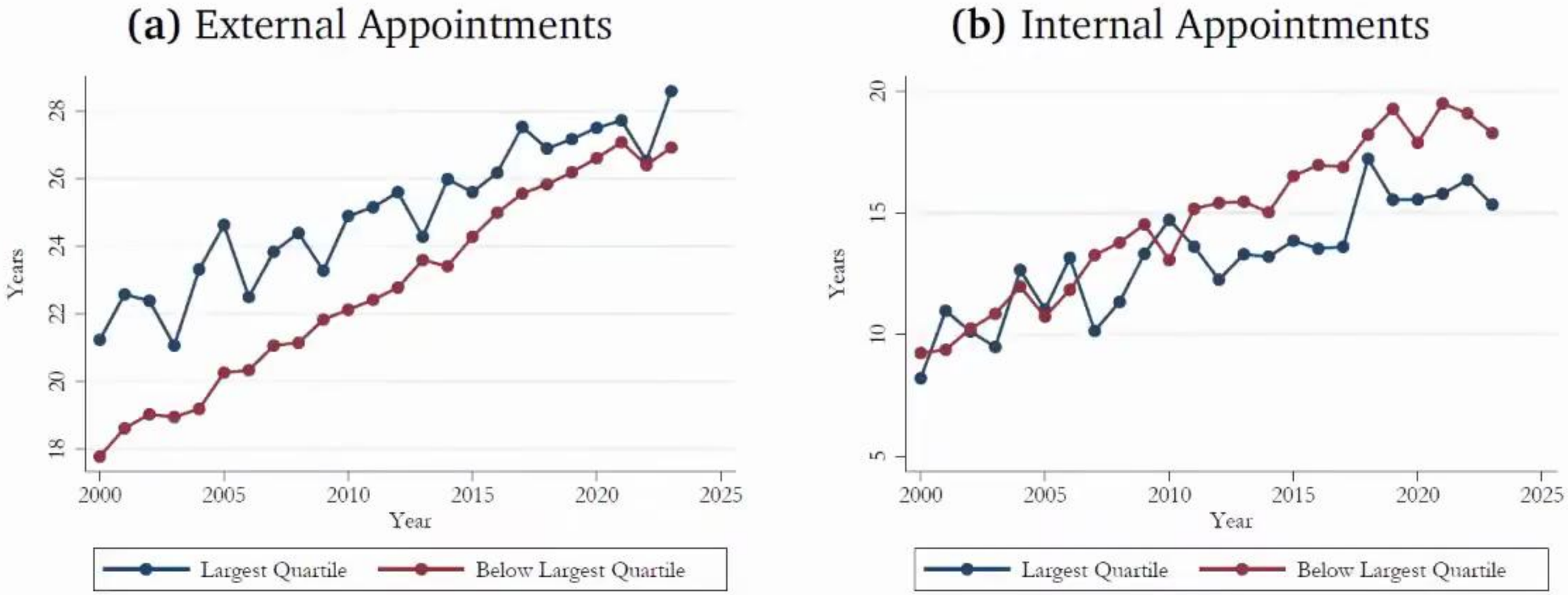


External vs. Internal Experience Prior to CEO Appointmen

Years of External vs. Internal Experience

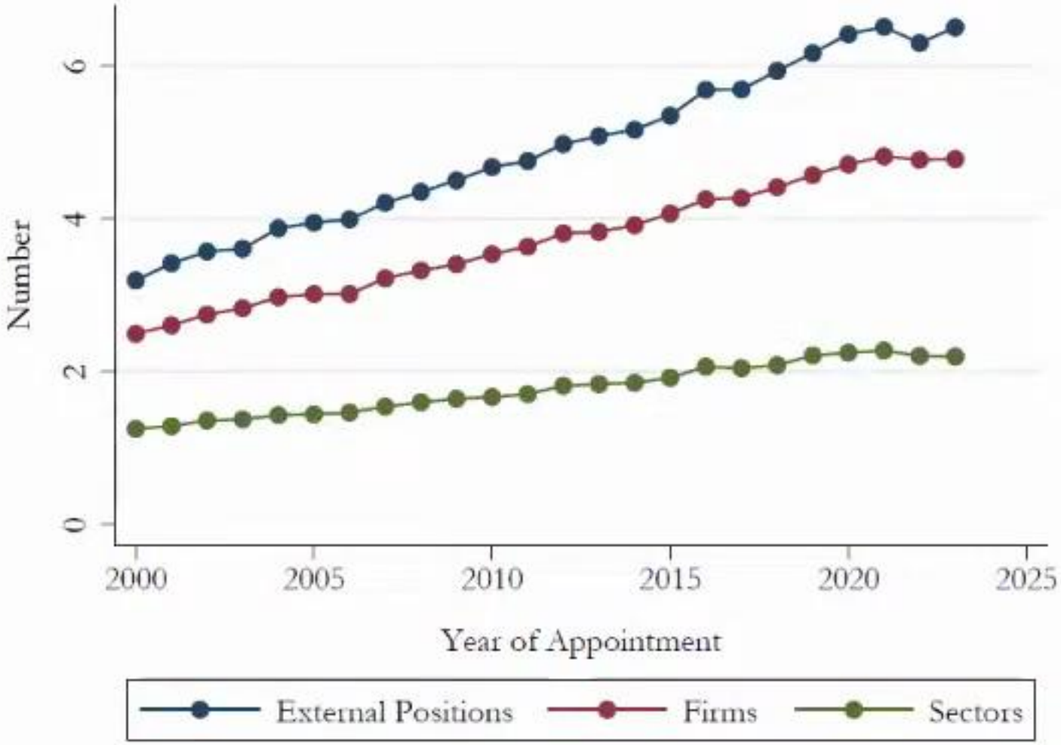


External Experience by Firm Size

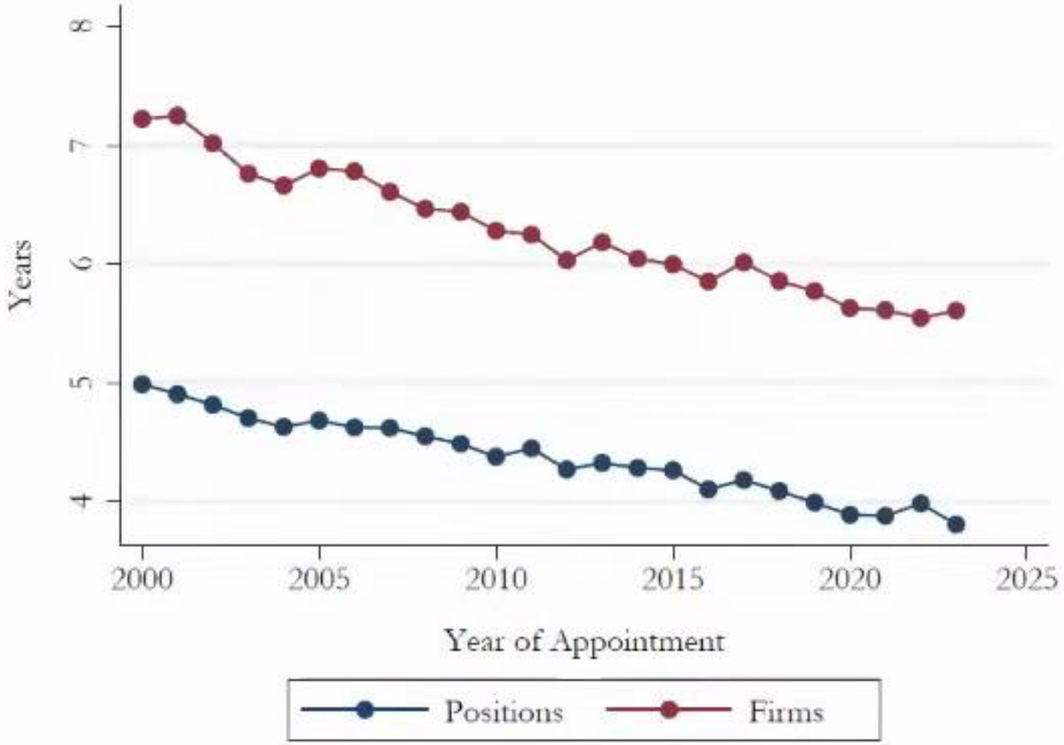


Work Experience Prior to CEO Appointment

(a) No. of Positions, Firms, Sectors



(b) Years per Position and Firm



Comparison Non-CEOs Extensive Margin

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Theoretical Framework

Assignment Model with Heterogeneous Workers & Firms

- Frictionless many-to-one matching model with transfers Details
- Workers with multidimensional skills: age-adjusted ability $x(a, t)$, experience $y(t)$
- Assortative matching $\rightarrow$ define firm's most productive worker as CEO

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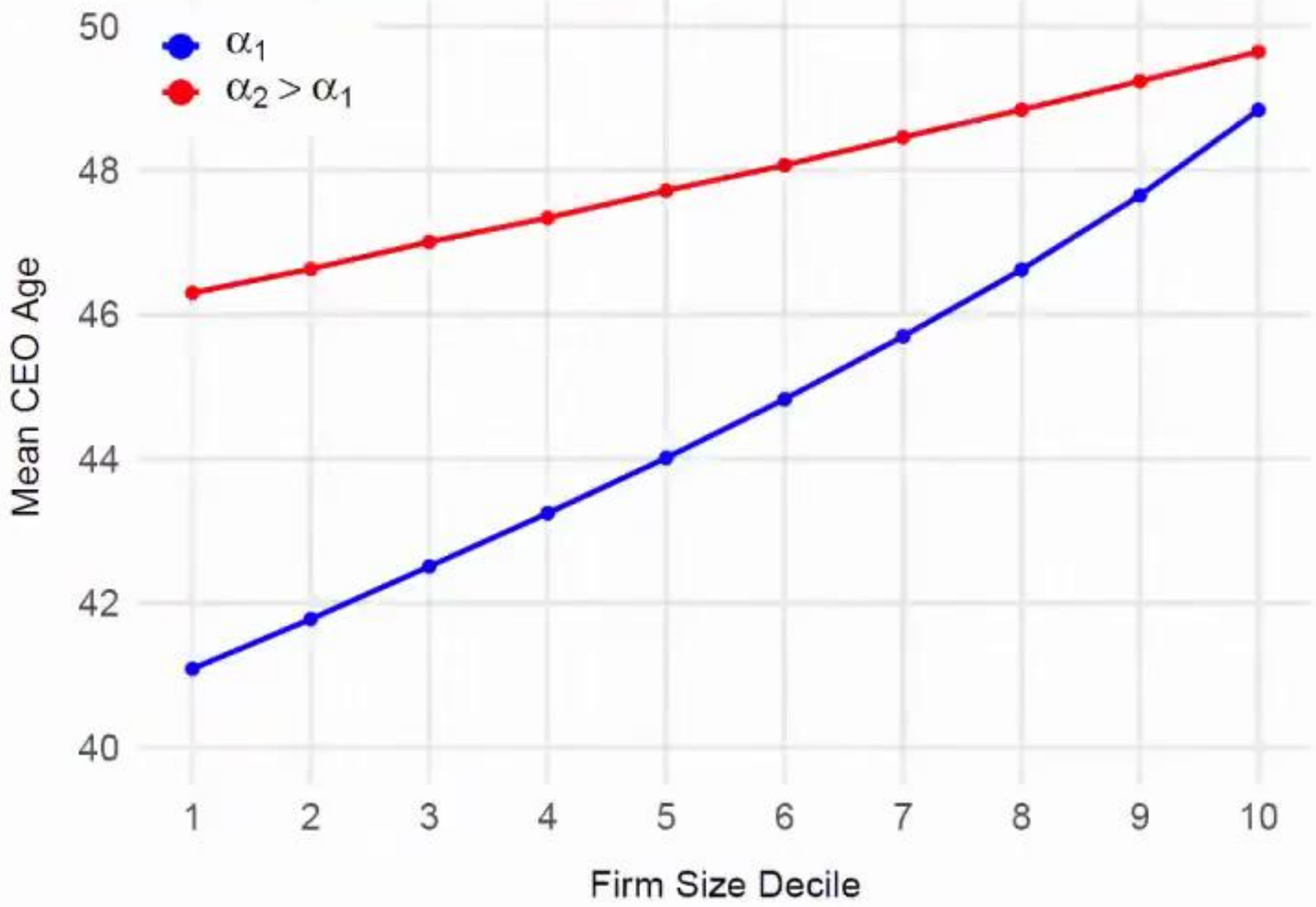
Key Results Assumptions

- CEOs of smaller (lower-productivity) firms are younger
- Weight on experience $\alpha \uparrow$
  - Age increase among all CEOs
  - Especially so at smaller firms

Illustrative CEO Age Profiles by Firm Size Decile



Notes: $\alpha_1 = 0.4$, $\alpha_2 = 0.8$. 100,000 workers with ability $a \sim U(0, 200)$, age $t \sim U(0, 50)$, $h(t) = -0.04t(t - 50)$, and $y(t) = -0.1t(t - 150)$. 1,000 firms with productivity $z \sim U(0, 1)$, capacity $m(z) = \lceil (1 + 5z)^2 \rceil$, and decay parameter $\beta = 0.08$. The simulation is repeated 500 times.

Demand Side: Changes in Skill Requirements

Identifying Changes in Skill Requirements

Hypothesis: demand for generalist CEOs ↑ (Decressin et al., 2023)

- **Secular increase in uncertainty** (Baker et al., 2016) Plot
- Business & spatial (organizational) diversification, trade-induced complexity Plots

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Resolving endogeneity concerns

- Use exogenous spatial variation in the availability of young generalists
- Study effect on CEO age at appointment in high- vs. low-uncertainty industries

Tying Generalist Skills to Age of Prospective CEOs

Source of variation: availability of top strategy consultants (McKinsey, BCG, Bain)

- Generalist skills through business education/MBAs and on-the-job training
- Generate value added as **substitutes for experience**

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Measurement: metro-area variation in **flight time to closest MBB office** Time Series

- Office locations across the US (hand-collected) Offices
- Quasi-random exposure via air route expansions (Bureau of Transport Statistics)

Rising Uncertainty Spurred Demand for Generalists

	CEO Age			
	(1)	(2)	(3)	(4)
Low Supply	0.251	-0.708		
	(0.274)	(0.893)		
Uncertainty	-0.494	-0.659**	-0.734*	
	(0.350)	(0.307)	(0.378)	
Low Supply × Uncertainty	0.477**	0.626***	0.477**	0.633***
	(0.183)	(0.232)	(0.212)	(0.212)
Observations	15,679	15,676	15,235	14,132
MSA fixed effects		✓		
Year fixed effects		✓		
MSA-Year fixed effects			✓	✓
Naics4 fixed effects			✓	
Naics4-Year fixed effects				✓

- 1 h ↑ in travel time at 1 σ above mean uncertainty ≈ 0.6 years older CEO
- Analogous results for other drivers of firm-level demand for generalists

By Firm Size | Other Outcomes | Event Study | Time-Varying Estimates | Robustness | Extensive Margin | Age Externality | MBB Experience

Causal Effects on Firm Performance—Two-Step Approach

Idea: use result on uncertainty as a first stage to study real effects

Consider firm outcome v_{jt} and CEO w in firm j, industry i, and year t

1. Estimate **firm-by-CEO fixed effects**

$$v_{jt} = \nu_{wj} + \phi_{it} + X_{jt} + \epsilon_{jt}$$

   - X_{jt}: log number of employees, firm age, and CEOs tenure at the firm
   - ν_{wj} captures CEO performance net of industry and firm-level covariates

2. Estimate effects of **CEO age at appointment on measure** $\hat{\nu}_{wj}$

$$\hat{\nu}_{wj} = \beta \widehat{Age}_{jt} + \gamma_{mt} + \phi_{it} + \epsilon_{jt}$$

   - Instrument Age_{jt} with $Low\ Supply_{mt} \times Uncertainty_{it}$

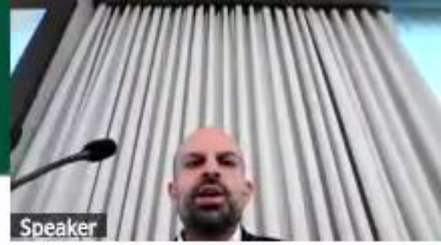
CEO Age and Firm-Level Outcomes—IV Regressions

	Emp. Gr.	R&D Gr.	Any Patent	Log(Patent)	Log(Comp. Patents)	Share Comp. Patent
	(1)	(2)	(3)	(4)	(5)	(6)
CEO Age	-0.017*	-0.036***	-0.016	0.039	-0.057**	-0.022***
	(0.010)	(0.011)	(0.016)	(0.038)	(0.023)	(0.008)
R^2	-0.230	-0.864	-0.165	-0.063	-0.357	-0.460
Observations	2,495	1,136	9,873	1,946	1,946	1,946
MSA-Year fixed effects	✓	✓	✓	✓	✓	✓
Naics4-Year fixed effects	✓	✓	✓	✓	✓	✓
Kleibergen-Paap F-Statistic	11.81	33.06	14.04	10.03	10.03	10.03

Correlations Business-Cycle Shocks Bootstrapped Standard Errors

Supply Side: Workers' Responses to Changed Skill Requirements

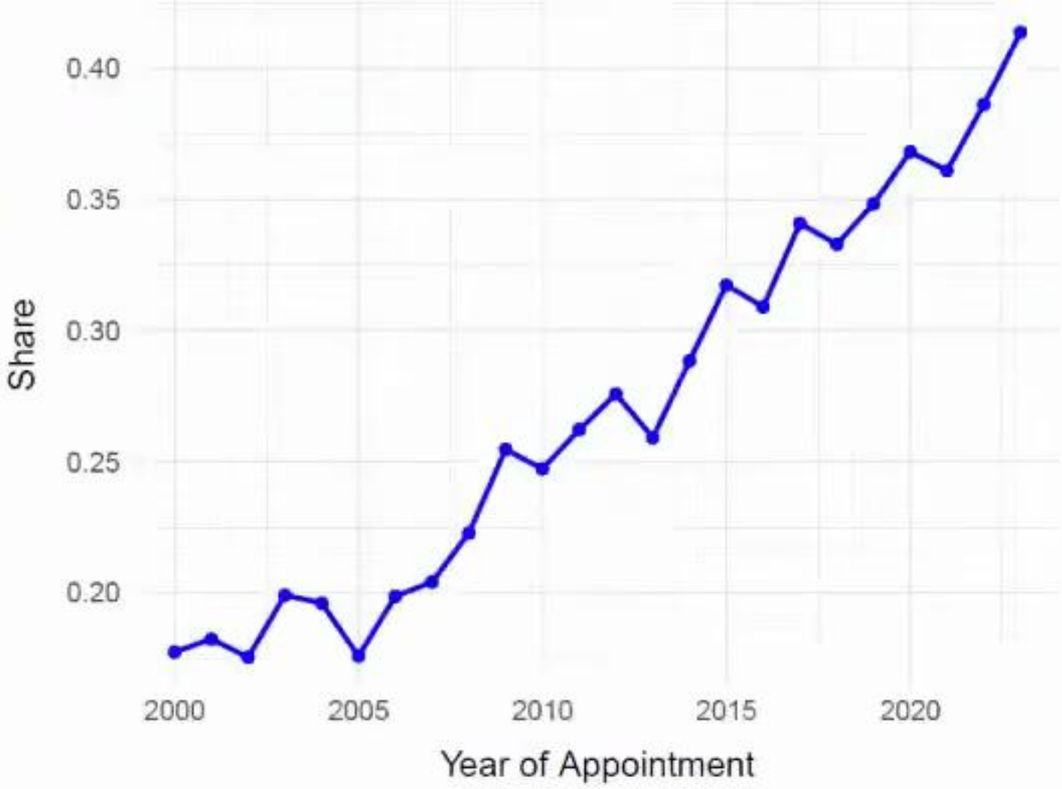
Executives' Decision to Acquire Generalist Skills



Main indicator: job switches across firms to less senior positions

- Downward switches more likely motivated by pursuit of generalist skills

Share of CEOs That Ever Moved Downwards Across Firms



Overall Switches

Exploiting Network Shocks To Establish Causality

Idea: Workers emulate successful career moves of connected peers

Difference-in-differences around information shocks on value of job mobility

- **Treatment:** former coworker becomes CEO
  - Within 10 years after leaving firm and having switched firms at least once
- **Treated group:** worked alongside the CEO
  - Same firm, metro area, and seniority level for at least 3 months
- **Control group:** same firm and seniority level, but worked in *different* metro area

Information Shocks Increase Voluntary Turnover

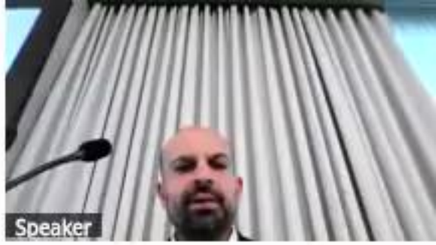
	Any Firm (1)	Upward (2)	Lateral (3)	Downward (4)	Any Sector (5)
Treat × Post	0.005***	-0.001	0.000	0.005***	
	(0.001)	(0.000)	(0.001)	(0.001)	
Observations	22,777,381	22,777,381	22,777,381	22,777,381	
Pre-Treatment Control Mean	0.094	0.025	0.029	0.041	
Individual fixed effects	✓	✓	✓	✓	
Firm-Seniority-Year fixed effects	✓	✓	✓	✓	
Different fixed effects	Results	Results	Results	Results	

- Firm switches ↑ by ≈ 5% during 10-year window after CEO appointment
- Downward switches ↑ by ≈ 13%

Event Study | Weighted Regressions | Summary Statistics

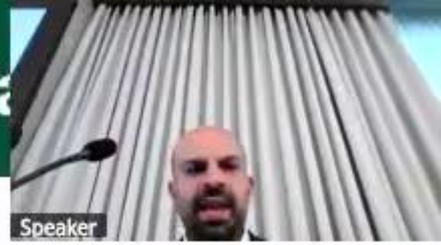
Comparative Statics

- **Results** are
  - driven by switches across job categories Table instead of within Table
  - stronger for coworkers with longer shared tenure Table Plot
  - similar for employees at seniority levels just above Table and below Table
  - driven by period since 2005 Table
  - stronger for high-mobility CEOs Table
  - stronger depending on direction of CEO switch Table
  - driven by workers in small firms Table Plot Time Variation in Trade-Off
  - driven by switches towards smaller firms Table
- Lower short-run **salary growth** in response to treatment Table



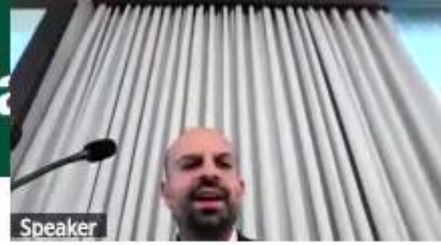
Decomposing Demand- vs. Supply-Side Factors

Quantifying Contributions of Demand- and Supply-Side Fa



- **Regress CEO age on different combinations of fixed effects** Details
- **Determine contributions based on R^2 of regressors** Details
  - Use Shapley values to deal with asymmetries from sequentially adding fixed effects

Quantifying Contributions of Demand- and Supply-Side Fa



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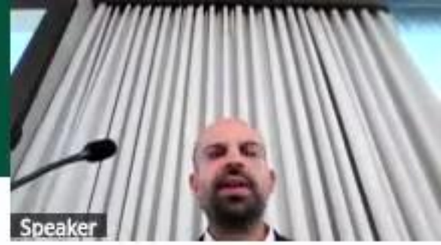
Factor	Interpretation	Contribution (SE)
Demand	Sectoral shifts in demand	37.1% (0.8)
Supply	Local availability of generalists ▶ labor market thickness ▶ executives' reaction to demand shifts ▶ executive training pipelines	28.4% (0.7)
Interaction	Supply-side frictions affect adjustment to demand	34.5% (0.9)

Full Table

Conclusion



- **Upward age trend pervasive among CEOs**
- **Demand-side shift towards generalist skills** to navigate fluctuations
- **Supply-side reaction:** executives strategically increase job mobility
- **Negative impact** on firm growth and (computing-related) innovation
- **Aggregate implications**
  - Firms' adaptability to long-term challenges ↓
  - Changes in life-cycle earnings patterns at the top